Ex. Nos. 6	Snort IDS/IPS to detect and prevent real time threats in TryHackMe Platform
Date :	

AIM:

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

SOFTWARE REQUIRED:

TryHackMe account, AttackBox or VPN, Snort, Nmap, Linux terminal. ***HOW TO START TRYHACKME***

1. Login to TryHackMe.

2. Search for "Snort" room.

3. Click Start Machine.

4. Click Start AttackBox.

5. Open Terminal in AttackBox.

6. Verify Snort:

snort -V

7. Test configuration:

sudo snort -T -c /etc/snort/snort.conf

Experiment 1 — ICMP (Ping) Detection (IDS) — line by line (with new terminal steps)

1. Open terminal and Check Snort:

   snort -V

2. Test Configuration

   sudo snort -T -c /etc/snort/snort.conf

3. Open rule File

   sudo nano /etc/snort/rules/local.rules

4. Add this rule:(ENSURE ONLY THIS RULE EXISTS)

   alert icmp any any -> any any (msg:"ICMP Ping Detected";
   sid:1000001; rev:1;)

5. Save and exit: Ctrl+O → Enter → Ctrl+X

6. Run Snort (IDS Mode)

sudo snort -c /etc/snort/snort.conf -i eth0

7. Generate Attack (New Terminal) using Open new terminal In AttackBox:

● Click + (New Tab) at top of terminal
OR

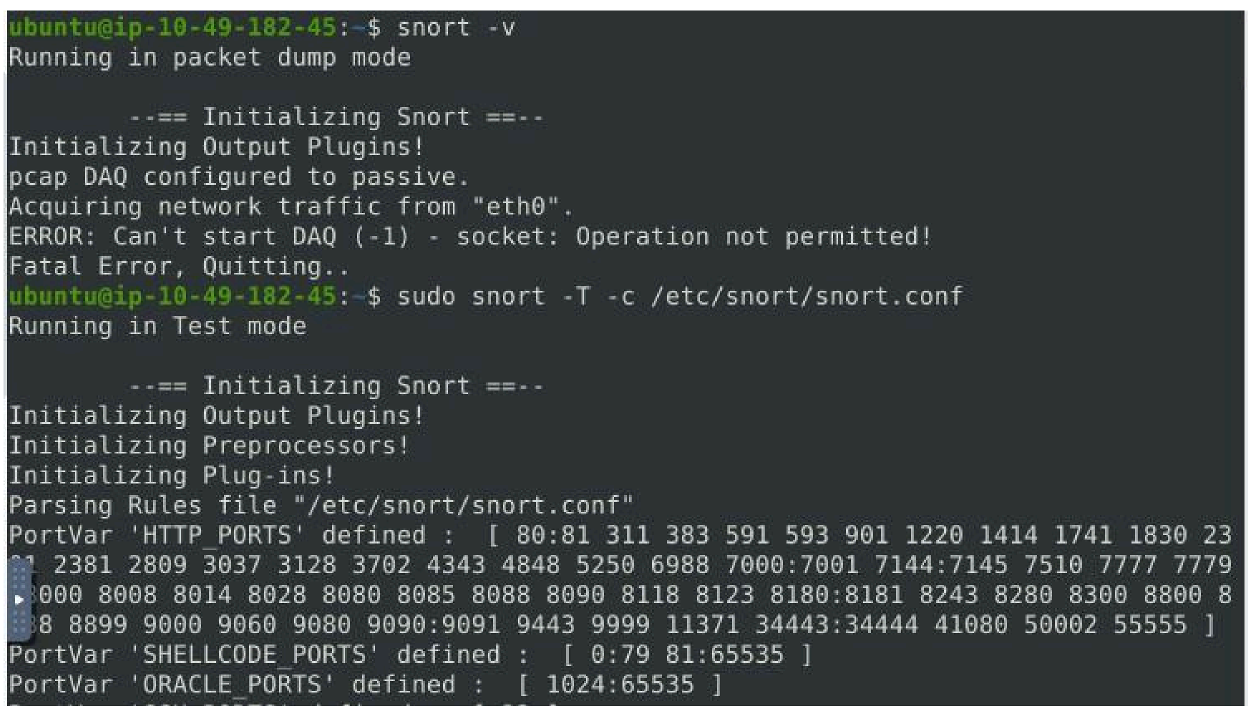
● Right-click terminal → New Tab

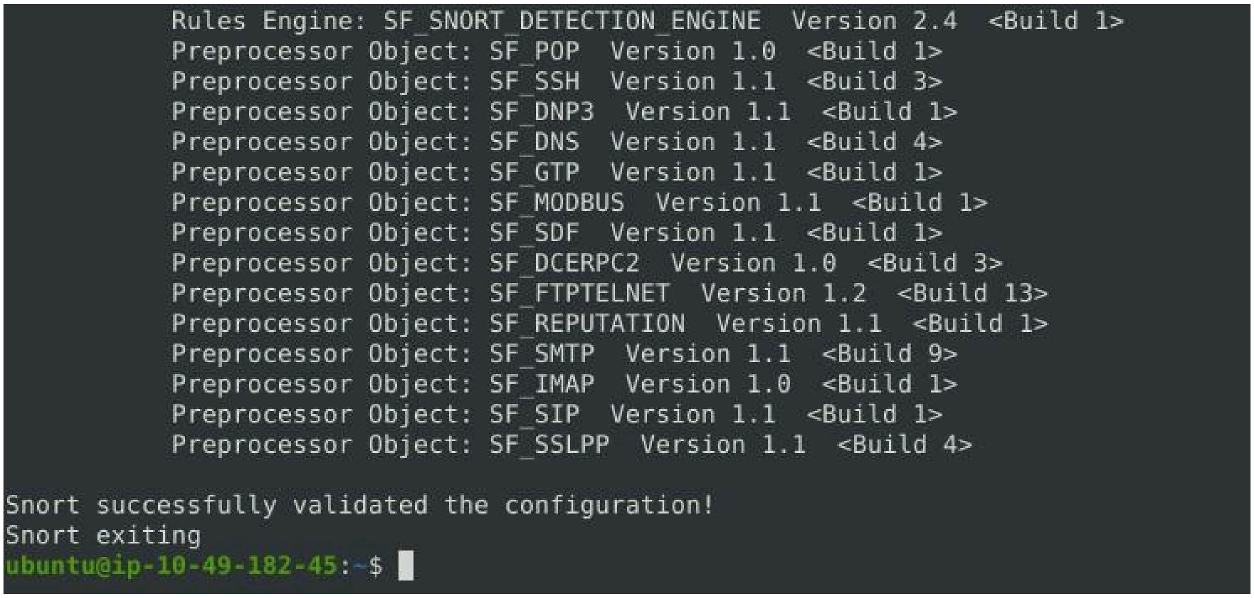
8. In the new terminal:(ATTACK BOX IP ADDRESS) ping 10.48.190.168

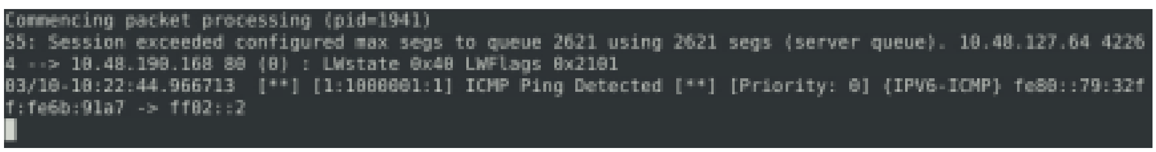
9. Go back to first terminal and View Alert → confirm alert shown ICMP Ping Detected

10.(Log check – optional):

sudo tail -f /var/log/snort/alert







Experiment 2 — Port Scan Detection — line by line (with new terminal steps)

1. Edit Rules File

   sudo nano /etc/snort/rules/local.rules

2. Add:

   alert tcp any any -> any any (msg:"TCP Port Scan Activity";
   sid:1000002; rev:1;)

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

   Ctrl + C

5. Restart Snort:

   sudo snort -A console -c /etc/snort/snort.conf -i eth0

6. Open new terminal tab

● Click + on terminal
OR

● Right-click → New Tab

7. In new terminal:

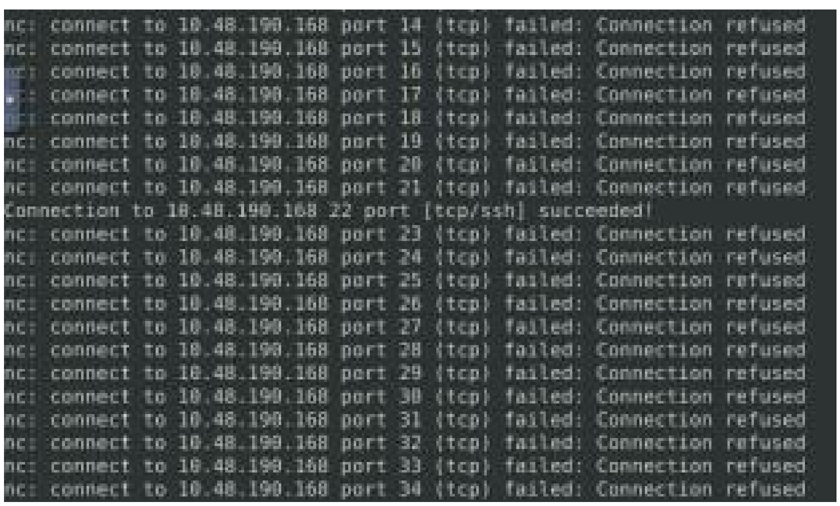
    for p in {1..100}; do nc -zv 10.48.190.168 $p; done

8. Go back to Snort terminal → confirm alert

9. (Optional log):

    sudo tail -f /var/log/snort/alert

```
03/10-11:03:14.096398  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.096939  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.098994  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.098369  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.099098  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.099989  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.101333  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.118446  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.118519  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
03/10-11:03:14.119767  [**] [1:1000002:1] TCP Port Scan Activity [**] [Priority: 0] {TCP} 10.48.127.64:4
2204 -> 10.48.190.168:80
```



```
nc: connect to 10.48.190.168 port 14 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 15 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 16 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 17 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 18 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 19 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 20 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 21 (tcp) failed: Connection refused
Connection to 10.48.190.168 22 port [tcp/ssh] succeeded!
nc: connect to 10.48.190.168 port 23 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 24 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 25 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 26 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 27 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 28 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 29 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 30 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 31 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 32 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 33 (tcp) failed: Connection refused
nc: connect to 10.48.190.168 port 34 (tcp) failed: Connection refused
```

Experiment 3 — IPS Blocking (Inline Mode) — line by line (with new terminal steps)

1. Change rule in Rule File

    sudo nano /etc/snort/rules/local.rules

2. Add:
    drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003; rev:1;)

3. Save: Ctrl+O → Enter → Ctrl+X

4. Stop Snort:

    Ctrl + C

5. Run inline:

    sudo snort -Q --daq afpacket -i eth0:eth0 -c /etc/snort/snort.conf 6. Open

new terminal tab

● Click +
OR

● Right-click → New Tab

7. In new terminal:

    ping 8.8.8.8

8. Expected result:
    No ping reply
    Alert shown in Snort terminal

9. (Optional log):

    sudo tail -f /var/log/snort/alert

If interface is not eth0

Run once:

ip a

Use the shown interface name instead of eth0.

RESULT:

Successfully detected and prevented attacks using Snort IDS/IPS.